

**PERFORMANCE EVALUATION OF A SOLAR INVERTER UNDER
VARIABLE LOAD CONDITIONS**

BY

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DEFINITION OF TERMS

Alternating current (AC): Electric current whose direction reverses periodically; the form of electricity used by most appliances.

Ambient temperature: The temperature of the air surrounding the equipment at the test location.

Apparent power: The product of the RMS voltage and the RMS current, measured in volt-amperes (VA).

CEC efficiency: A weighted efficiency measure developed by the California Energy Commission that combines the efficiencies at several load levels using fixed weighting factors.

Conduction loss: Power lost as heat when current flows through the resistance of the components of the inverter.

Direct current (DC): Electric current that flows in one direction only, as produced by PV modules and batteries.

Euro (European) efficiency: A weighted efficiency measure that combines the efficiencies at several load levels using fixed weighting factors derived for Central European conditions.

Full (rated) load: A load equal to the manufacturer's rated output power of the inverter, which is 1000 W for the inverter tested.

Insulated Gate Bipolar Transistor (IGBT): A semiconductor switching device commonly used in higher-power inverters.

Inverter efficiency: The ratio of the AC output power of the inverter to the DC input power supplied to it, expressed as a percentage.

Load: Any electrical device, or group of devices, that consumes power from the inverter output.

Maximum Power Point Tracking (MPPT): A control technique that keeps a PV source operating at the point where the product of its voltage and current is greatest.

Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET): A semiconductor switching device that switches at high frequency and is commonly used in lower-power inverters.

No-load voltage: The output voltage of the inverter when it is switched on with no load connected.

Photovoltaic (PV) system: A system that converts solar radiation directly into electrical energy using photovoltaic cells and associated components.

Power factor (PF): The ratio of real (active) power to apparent power; it is close to 1 for a resistive load.

Power loss: The difference between the DC input power and the AC output power of the inverter, dissipated mainly as heat.

Power quality: The characteristics of an electrical power supply, including voltage stability, frequency consistency, and waveform quality.

Pulse Width Modulation (PWM): A control method that varies the width of switching pulses so that the filtered output approximates a sinusoidal waveform.

Real (active) power: The power that is actually converted into useful work or heat by the load, measured in watts (W).

Resistive load bank: An adjustable set of resistive loads used to set the inverter load at known percentages of its rated capacity.

Solar inverter: An electronic device that converts DC electricity from solar panels or batteries into AC electricity suitable for electrical appliances.

State of charge (SoC): The amount of charge remaining in a battery, expressed as a percentage of its full capacity.

Switching loss: Power lost in the semiconductor switches of the inverter each time they turn on and off.

Total Harmonic Distortion (THD): A measure of the distortion of a waveform caused by harmonics, relative to the fundamental component.

Variable load condition: A condition in which the load demanded from the inverter is changed; in this study, in steps of 20% of the rated capacity.

Voltage drop: The percentage fall in output voltage between no-load and a given partial-load level.

Voltage regulation: The percentage change in the output voltage of the inverter between no-load and full-load conditions, used as a measure of voltage stability.

ABSTRACT

Solar inverters convert the direct current (DC) from batteries and photovoltaic panels into the alternating current (AC) that most appliances require, but their performance changes with the load they supply, whereas manufacturers usually quote a single peak or rated-condition value. This project evaluated the performance of a 1000 W solar inverter under variable load conditions. A quantitative experimental design was adopted. The inverter was powered from a 12 V deep-cycle battery and loaded with a controlled resistive load bank at 20%, 40%, 60%, 80%, and 100% of its rated capacity. At each level, the DC input voltage and current, the AC output voltage, current, and power, the power factor, and the case temperature were measured with a multimeter, a clamp meter, and a wattmeter, and the mean of three repeated runs was analysed. Efficiency was lowest at 20% load (82.0%), reached a peak of 93.0% at 60% load, and fell to 89.0% at full load. The power loss was almost constant at 44–45 W up to 60% load and then rose to 123.6 W at full load, which explains this pattern. The AC output voltage fell from 230.5 V at 20% load to 225.8 V at full load, giving a voltage regulation of 2.75% relative to the no-load voltage of 232.0 V; the output current rose almost in proportion to the load, from 0.87 A to 4.43 A; the power factor remained between 0.99 and 1.00; and the case temperature rose from 31°C to 53°C. All three null hypotheses were rejected under the stated decision rule. The study concludes that, for the inverter tested, a single peak efficiency figure does not describe real performance, and it recommends operating such inverters in the mid-range load region, about 40% to 80% of rated capacity, with adequate cooling near full load. The main limitations are the use of a single inverter, resistive loads only, a short test duration, the ascending order of the loads, and the absence of an inferential test on the individual readings.

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

The increasing global demand for sustainable and reliable energy has accelerated the adoption of renewable energy technologies, particularly solar photovoltaic (PV) systems. Solar energy is abundant, environmentally friendly, and capable of supplying electricity in both grid-connected and remote locations. However, PV panels produce direct current (DC), whereas most domestic, commercial, and industrial appliances operate on alternating current (AC). Solar inverters are therefore essential components of PV systems because they convert the DC electricity from the panels or batteries into usable AC electricity (Messenger & Abtahi, 2010).

Beyond DC-to-AC conversion, modern inverters regulate output voltage and frequency, extract the maximum available energy from PV modules through Maximum Power Point Tracking (MPPT), and provide system protection (Teodorescu et al., 2011). The performance of an inverter therefore determines both the amount and the quality of the electrical energy delivered to consumers, which makes its evaluation an important part of PV system design and operation.

The operating conditions of an inverter are not constant in practice. Variations in load demand, solar radiation, and ambient temperature can affect conversion efficiency, output voltage stability, current characteristics, power quality, and thermal behaviour. An inverter operating at light load may show different efficiency characteristics from one operating near its rated capacity (King et al., 2007; Chmielowiec et al., 2022). Evaluating inverter performance under different load conditions therefore provides valuable information about its suitability for practical applications.

Solar PV has expanded rapidly because of technological improvements, falling costs, and growing concern about the environmental effects of fossil fuels. The International Energy Agency (IEA, 2024) expects solar PV to account for almost 80%

of the growth in renewable electricity worldwide through 2030. This expansion makes it important that supporting components, particularly inverters, operate efficiently and reliably.

Reliability matters most in stand-alone and weak-grid settings, where a PV system is often the primary or only source of electricity. In Nigeria, for example, more than 85 million people lacked access to electricity in 2021, and households that are connected to the grid often fill gaps in an unreliable supply with costly and polluting generator sets (World Bank, 2023). In such settings, poor inverter performance directly causes energy losses and unstable power delivery.

Performance evaluation of a solar inverter involves measuring and analysing parameters such as input power, output power, efficiency, voltage regulation, and current response under controlled conditions. Such evaluation reveals the strengths and limitations of an inverter and helps in selecting appropriate ratings for specific applications. Manufacturers state rated specifications, but a single rated or peak figure does not show how an inverter behaves across its load range. This study therefore evaluates a solar inverter under different load conditions to determine how changes in demand affect its efficiency, stability, and overall operational behaviour.

1.2 Statement of the Problem

The growing adoption of solar PV systems has increased the need for efficient and reliable inverters. Although inverters are designed to provide stable conversion of DC power into usable AC electricity, their performance varies with operating conditions. Variations in electrical load demand can affect conversion efficiency, output voltage stability, and power losses.

In practice, inverters operate under continuously changing loads rather than at a fixed rated condition, yet manufacturer specifications commonly quote a single peak or rated-condition figure that does not show how performance changes across the load range. A poor understanding of inverter behaviour under variable load can lead to

improper system sizing, reduced efficiency, overheating, and increased maintenance costs. There is therefore a need to evaluate experimentally how the performance of a solar inverter changes under different load conditions, and this study addresses that need.

1.3 Aim and Objectives of the Study

Aim of the Study

The aim of this study is to evaluate the performance of a solar inverter under variable load conditions by investigating its efficiency, output characteristics, and operational stability.

Objectives of the Study

The specific objectives of this study are to:

1. determine the input and output power of a solar inverter at different load levels;
2. evaluate the conversion efficiency of the inverter at varying electrical loads;
3. examine the effect of load variation on the inverter's output voltage, output current, and power factor;
4. assess the operational stability of the inverter, in terms of voltage regulation and operating temperature, across the load levels tested; and
5. identify the relationship between load demand and the overall performance of the inverter.

1.4 Research Questions

This study seeks to answer the following research questions:

1. How does variation in electrical load affect the output performance of a solar inverter?
2. What is the efficiency of the inverter at different load levels?
3. How does changing load demand influence the output voltage, output current, and power factor of the inverter?
4. How stable is the inverter, in terms of output voltage and operating temperature, under variable load conditions?

5. What relationship exists between load variation and the overall performance of the inverter?

1.5 Research Hypotheses

In line with the objectives and research questions, the following null hypotheses (H_0) and their alternatives (H_1) are formulated to guide the analysis of results:

H_{01} : There is no significant difference in the conversion efficiency of the inverter across the load levels tested.

H_{11} : There is a significant difference in the conversion efficiency of the inverter across the load levels tested.

H_{02} : Variation in electrical load has no significant effect on the output voltage of the inverter.

H_{12} : Variation in electrical load has a significant effect on the output voltage of the inverter.

H_{03} : There is no significant relationship between load demand (as a percentage of rated capacity) and the overall performance (efficiency and voltage regulation) of the inverter.

H_{13} : There is a significant relationship between load demand (as a percentage of rated capacity) and the overall performance (efficiency and voltage regulation) of the inverter.

The decision rule used to accept or reject each null hypothesis is stated in Section 3.9.

1.6 Significance of the Study

The study provides useful information on how changes in electrical demand influence the operation of a solar inverter. For solar system designers and engineers, this information can assist in inverter selection, system sizing, and the optimisation of PV

installations, since operating an inverter in its more efficient load range can help reduce energy losses.

The research is also useful to users of solar PV systems because it draws attention to the importance of operating inverters within suitable load ranges. It contributes to academic knowledge by providing experimental data on inverter performance that students, researchers, and future studies on renewable energy systems and power electronics can use as a reference. Finally, the findings may support wider adoption of solar energy by highlighting the factors that influence inverter performance, particularly in regions where reliable electricity supply remains a challenge.

1.7 Scope of the Study

This study focuses on the performance evaluation of a solar inverter under different electrical load conditions. It examines how variations in load demand affect the inverter's output voltage, output current, power output, power factor, conversion efficiency, and operating temperature.

The study is limited to the experimental assessment of a single commercially available 1000 W solar inverter under controlled laboratory conditions, using resistive loads set at 20%, 40%, 60%, 80%, and 100% of the inverter's rated capacity. It does not cover the design and construction of a new inverter, the manufacture of PV modules or batteries, or the analysis of the national electricity grid.

1.8 Limitations of the Study

Like every experimental study, this research is subject to certain limitations. The accuracy of the results depends on the precision and calibration of the available measuring instruments. The study also evaluates only one inverter under controlled conditions using resistive loads, so the results may not fully represent other inverter models, capacities, or load types (such as inductive or non-linear loads), or conditions such as varying solar radiation, ambient temperature, and long-term field use.

In addition, fluctuations in the power supply, environmental factors, and the short duration of the tests may have affected the measurements. The procedures described in Chapter Three were adopted to minimise these effects, and the specific limitations of the methodology are discussed in Section 3.11.

1.9 Operational Definition of Terms

The following terms are defined as they apply to this study:

Solar Photovoltaic (PV) System: A system that converts solar radiation directly into electrical energy using photovoltaic cells and associated components.

Solar Inverter: An electronic device that converts DC electricity from solar panels or batteries into AC electricity suitable for electrical appliances.

Variable Load Condition: A condition in which the load demanded from the inverter is changed; in this study, it is changed in defined steps of 20% of the rated capacity.

Load: Any electrical device, or group of devices, that consumes power from the inverter output.

Inverter Efficiency: The ratio of the AC output power produced by the inverter to the DC input power supplied to it, expressed as a percentage.

Voltage Regulation: The percentage change in the inverter's output voltage between no-load and full-load conditions, used as a measure of output voltage stability.

Voltage Drop: The same percentage change, calculated between no-load and any partial-load level; it equals the voltage regulation at 100% load.

Power Factor: The ratio of real (active) output power to apparent output power.

Power Quality: The characteristics of an electrical power supply, including voltage stability, frequency consistency, and waveform quality.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The increasing demand for clean energy has driven significant growth in solar photovoltaic (PV) technology, in which the solar inverter plays a critical role by converting the DC electricity generated by the panels, or stored in batteries, into the AC electricity required by most appliances. The performance of a PV system therefore depends greatly on the efficiency and reliability of its inverter, since conversion losses directly reduce the usable power delivered to consumers.

In practice, inverters are exposed to constantly changing loads. Inverter efficiency is not constant but changes with loading level: fixed internal losses reduce efficiency at very low loads (King et al., 2007), while high loads increase conduction losses, switching losses, and thermal stress (Mohan et al., 2003; Wang et al., 2013). This chapter reviews literature on PV systems and their components, solar inverters and their classification and topologies, performance parameters and testing standards, the effect of variable load, relevant theories, the conceptual framework of the study, and previous empirical studies.

2.2 Solar Photovoltaic Energy Systems

PV systems convert sunlight directly into electricity through the photovoltaic effect, in which semiconductor materials absorb solar radiation and release electrons to produce an electric current. A typical system consists of PV modules, power-conversion devices, protection components, energy storage (where required), and monitoring systems, and is classified as grid-connected, stand-alone, or hybrid depending on its relationship with the utility network (Messenger & Abtahi, 2010). The electricity generated depends on solar radiation, temperature, module efficiency, and system design, but the effectiveness of the whole system also depends strongly on the inverter, which converts and regulates the power for AC appliances.

2.2.1 Global Growth of Solar Photovoltaic Energy

The IEA (2024) forecasts that solar PV and wind will account for 95% of all renewable capacity additions through 2030, with solar PV alone providing almost 80% of the growth in renewable electricity worldwide. This growth is attributed to falling equipment costs and wide social acceptance, and distributed applications (residential, commercial, industrial, and off-grid) are expected to make up nearly 40% of PV expansion (IEA, 2024). Such growth increases the demand for reliable supporting technologies, particularly inverters and energy storage.

2.2.2 Solar Energy in Stand-Alone and Weak-Grid Applications

In stand-alone and weak-grid settings, the utility supply is either unavailable or too unreliable to depend on exclusively. In Nigeria, for instance, over 85 million people lacked electricity access in 2021, and grid-connected households have faced unreliable and insufficient supply that is often supplemented with petrol and diesel generators; the Nigeria Electrification Project alone distributed over one million solar home systems and established 125 mini-grids (World Bank, 2023). Adesina et al. (2023) designed and tested a 3.5 kVA off-grid inverter system for a Nigerian household, which supplied a daily load of 11,260 Wh for about 24 hours without interruption.

In such systems the inverter's performance is particularly critical. Because there is no grid supply to compensate for it, any inefficiency, instability, or failure at the inverter stage immediately reduces the electricity available to the end user. This makes the evaluation of the inverter under realistic loading important.

2.3 Components of a Photovoltaic System

A complete PV system consists of several components that work together, and the performance of the inverter cannot be fully appreciated without understanding the components alongside which it operates (Messenger & Abtahi, 2010). PV modules are the primary energy-generating components, converting solar radiation into DC electricity, and are connected in series and parallel combinations to form arrays that

provide the required voltage and current. In systems with battery storage, a charge controller regulates the flow of DC power between the array and the battery to prevent overcharging; charge controllers are commonly based on Pulse Width Modulation (PWM) or Maximum Power Point Tracking (MPPT), and MPPT designs generally extract more power from the array under varying irradiance (Esrām & Chapman, 2007).

Batteries store DC electricity generated during periods of high solar radiation for use when generation is low or unavailable. Lead-acid batteries remain widely used in stand-alone systems because of their low cost, while lithium-ion batteries are increasingly adopted where higher energy density and longer cycle life are needed (Messenger & Abtahi, 2010). The state of charge and voltage characteristics of the battery influence the stability of the DC input that the inverter receives, which is why the battery was fully charged to the same state before each run in this study (Chapter Three). Fuses, circuit breakers, and surge protection devices protect the system and the connected loads from over-current, short-circuit, and surge faults, while monitoring devices allow voltage, current, power, and energy to be observed and recorded.

Within this arrangement, the inverter occupies a uniquely important position because it is the stage at which the DC electricity produced and stored elsewhere in the system is converted into the AC form required by end-use appliances; a fault or inefficiency at this stage affects all of the energy generated upstream of it.

2.4 Solar Inverters

A solar inverter is a power-electronic device that converts DC electricity into AC electricity suitable for electrical appliances, serving as the link between the generation or storage source and the load (Teodorescu et al., 2011). Beyond basic conversion, modern inverters regulate output voltage and frequency, maintain power quality, provide protection, and monitor system operation. In grid-connected and hybrid inverters, MPPT enables the inverter to continuously adjust its operating point to obtain the maximum available power from the PV modules, whose output changes with solar

radiation and temperature (Esram & Chapman, 2007). Inverters are also equipped with over-voltage, over-current, short-circuit, and thermal protection against abnormal operating conditions.

The performance of a solar inverter is influenced by its design, semiconductor switching devices, control technique, operating temperature, and electrical loading (Teodorescu et al., 2011). Because an inverter rarely operates under constant load in practice, changes in demand cause variations in efficiency, output voltage stability, current characteristics, and thermal behaviour, and these can differ from the values given in manufacturer specifications, which are normally stated at rated conditions.

2.5 Types of Solar Inverters

Solar inverters are categorised by design, operating mode, capacity, and intended application (Kjær et al., 2005; Teodorescu et al., 2011). **Stand-alone (off-grid) inverters** operate independently of the utility network and are commonly combined with batteries, which makes them suitable for rural electrification and back-up power; since there is no external grid, they must maintain stable output voltage and frequency on their own. **Grid-connected inverters** operate alongside the utility network and must synchronise continuously with its voltage, frequency, and phase, and their power-quality performance is important because poor operation can introduce voltage fluctuations and harmonic distortion into the network. **Hybrid inverters** combine solar generation, battery storage, and grid interaction in one system, which gives flexibility in managing energy flow between panels, batteries, loads, and the grid.

By configuration, **string inverters** connect several panels in series to a single unit and offer a practical balance of cost and efficiency, although shading on some panels can reduce performance; **micro-inverters** are installed on individual panels so that each module operates at its own maximum power point, at a higher cost; and **central inverters** are high-capacity units used in large installations, which expose a larger part of the system to disruption if the unit fails (Messenger & Abtahi, 2010). The

unit evaluated in this study is a small-scale, battery-fed, stand-alone inverter of the type described above.

2.6 Inverter Topologies and Switching Devices

Beyond the functional classification in Section 2.5, inverters differ in their internal circuit topology and semiconductor switching devices, which affect efficiency, cost, and behaviour under variable load. **Transformer-based** inverters use a low- or high-frequency transformer for galvanic isolation between the DC and AC sides, which improves safety and reduces some types of leakage current, but the transformer adds weight, cost, and conduction losses. **Transformerless** inverters omit it and generally achieve higher efficiency, but they need more careful design to meet leakage-current and safety requirements (Kjær et al., 2005).

The single-phase H-bridge (full-bridge) topology, in which four semiconductor switches alternately reverse the polarity of the DC input, is a common basic configuration for producing an AC output in small and medium-sized inverters, while multilevel topologies synthesise the output from several voltage levels to reduce harmonic distortion and switching losses at the cost of more complex control (Mohan et al., 2003; Teodorescu et al., 2011). The switching action is performed by Insulated Gate Bipolar Transistors (IGBTs) or Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs). IGBTs are generally preferred in higher-power inverters because they handle larger currents and voltages with low conduction losses, whereas MOSFETs switch at higher frequencies with lower switching losses and suit lower-power inverters (Mohan et al., 2003). The choice of device affects how an inverter's losses are distributed across load levels, which is relevant to the load-dependent efficiency investigated here. The internal topology and switching devices of the unit tested were not examined in this study.

2.7 Inverter Performance Parameters

The performance of a solar inverter is described by several electrical and operational parameters that indicate its efficiency, reliability, and ability to deliver stable power: conversion efficiency, output voltage stability, output current, power quality, total harmonic distortion (THD), and thermal behaviour (Messenger & Abtahi, 2010).

Inverter Efficiency

Inverter efficiency is the ability of the inverter to convert DC input power into useful AC output power. It is the ratio of AC output power to DC input power, expressed as a percentage, as in Equation 2.1:

$$\text{Efficiency, } \eta (\%) = (P_{\text{out}} \div P_{\text{in}}) \times 100 \quad (2.1)$$

where η is the percentage efficiency, P_{out} is the AC power at the inverter output, and P_{in} is the DC power supplied to the inverter input.

Efficiency is not constant; it depends on load level, input voltage, switching losses, and operating temperature. Empirical inverter models include a self-consumption term, a loss that exists even at very small output (King et al., 2007), so at very low loads this fixed loss is a large share of the power handled and efficiency is reduced, while at high loads the growth of current-dependent conduction and switching losses can reduce it again (Mohan et al., 2003). In a laboratory study of 26 commercial PV inverters, Chmielowiec et al. (2022) measured conversion efficiency at generation levels from 25% to 100% of rated power. Evaluating efficiency at several load levels therefore gives a more realistic picture than testing at rated conditions alone.

Output Voltage Stability and Voltage Regulation

Output voltage stability is the ability of an inverter to maintain a consistent AC voltage despite changes in load demand, since significant fluctuations may reduce equipment performance or cause damage. It is commonly quantified using the percentage voltage regulation, defined in Equation 2.2:

$$\text{Voltage Regulation, VR (\%)} = [(V_{nl} - V_{fl}) \div V_{fl}] \times 100 \quad (2.2)$$

where V_{nl} is the output voltage at no load and V_{fl} is the output voltage at full (rated) load. A smaller VR indicates better stability. In this study the same expression, with the output voltage at a partial load level in place of V_{fl} , is called the voltage drop (VD) at that level, so that VR is the voltage drop at 100% load.

Weighted (Real-World) Efficiency Measures

The simple efficiency relation in Equation 2.1 describes performance at a single load point, and peak-efficiency figures are reached only briefly in real operation, so weighted measures have been developed. Two are widely reported: the “European efficiency” and the “CEC efficiency” (California Energy Commission). Both combine efficiencies measured at several load points using fixed weighting factors that reflect the time an inverter is assumed to spend at each level (Krauter & Bendfeld, 2024), as in Equations 2.3 and 2.4:

$$\eta_{\text{Euro}} = 0.03\eta_{5\%} + 0.06\eta_{10\%} + 0.13\eta_{20\%} + 0.10\eta_{30\%} + 0.48\eta_{50\%} + 0.20\eta_{100\%} \quad (2.3)$$

$$\eta_{\text{CEC}} = 0.04\eta_{10\%} + 0.05\eta_{20\%} + 0.12\eta_{30\%} + 0.21\eta_{50\%} + 0.53\eta_{75\%} + 0.05\eta_{100\%} \quad (2.4)$$

where each η term is the efficiency (Equation 2.1) at the load level in its subscript. The Euro formula, developed for Central European irradiance, weights the 50% load point most heavily, while the CEC formula, developed for the higher irradiance of California, places most weight on the 50% and 75% points. A full weighted calculation needs efficiency data at load points not all tested in this study, but the underlying principle, that partial-load efficiency matters as much as full-load efficiency, motivates the choice of five load levels (20% to 100%) rather than a single test at rated capacity.

Output Current, Power Quality, and Thermal Performance

The output current of an inverter changes with electrical demand, and abnormal current behaviour may indicate overloading or inefficient conversion. Power quality describes the characteristics of the supplied power, including waveform shape and THD; switching operations can introduce harmonics that raise THD and cause

overheating in connected equipment. Thermal performance is also important because heat produced by internal losses increases with load, and the reliability of power-electronic components is closely related to their operating stresses, including temperature (Wang et al., 2013).

2.8 Testing Standards for Solar Inverter Performance Evaluation

Inverter performance is not evaluated arbitrarily; international standards define recognised procedures for testing efficiency, safety, and grid behaviour. Table 2.1 lists selected standards relevant to this study, which is a small-scale academic evaluation rather than a formal certification test.

Table 2.1: Selected International Standards Relevant to Solar Inverter Testing

Standard	Scope
IEC 62109-1 / -2 (IEC, 2010; 2011)	Safety requirements for power converters used in PV power systems: general requirements (Part 1) and particular requirements for inverters (Part 2).
IEC 61683 (IEC, 1999)	Procedure for measuring the efficiency of power conditioners (inverters) in PV systems, including the load points at which efficiency is measured.
EN 50530 (CENELEC, 2010)	Method for testing the overall efficiency, including MPPT efficiency, of grid-connected PV inverters.
IEEE 1547 (IEEE, 2018)	Requirements for interconnecting distributed energy resources, including PV inverters, with electric power systems.

IEC 61683 requires efficiency to be measured at several defined load points rather than at a single rated point; for stand-alone inverters these range from 5% up to 120% of rated power (IEC, 1999). The five-level scheme used in this study (20% to 100%) applies the same principle of measuring at several defined load levels, but the levels are not those specified by the standard, and the study is not a formal standard test. Although it does not perform a safety or grid-interconnection test, the quantities recorded here (voltage, current, power, and efficiency at stated load levels) are of the kind defined by these standards.

2.9 Effect of Variable Load Conditions on Solar Inverter Performance

In practical applications, electrical loads are rarely constant, so efficiency and stability are not uniform across all loading levels. Loading is broadly grouped into light, medium, and heavy conditions. At light load, the fixed power consumed by control circuits and cooling represents a larger proportion of the power converted, which reduces efficiency. Medium loads generally place the inverter closer to its optimal efficiency range. At heavy load, close to rated capacity, current flow increases, and with it conduction losses, switching losses, and heat generation; if cooling is inadequate, the inverter may suffer reduced efficiency, thermal-protection activation, or accelerated component ageing (Mohan et al., 2003; Wang et al., 2013).

Experimental studies show that inverter performance depends on both the magnitude and the type of load. Chmielowiec et al. (2022) found that the efficiency indicators of the 26 inverters they tested depended on the operating point, and Az Zahro et al. (2025), testing a 5 kW high-frequency inverter with a resistive load bank of up to 1000 W, found that efficiency remained stable under purely resistive loads but fluctuated sharply when inductive loads were added, because of inrush currents. Vaicys et al. (2022) reported, from simulation compared with data from a conventional inverter, that a three-phase dual-system inverter could achieve a weighted efficiency 0.22–0.27% higher, showing that inverter configuration also matters.

Load variation similarly affects output current and thermal behaviour. As demand increases, the inverter must supply more current, and a well-designed inverter should regulate its output voltage despite this, although excessive loading may cause voltage drops and waveform distortion, while the accompanying rise in semiconductor stress increases heat generation. Testing under variable load therefore gives a more realistic assessment than a single rating, and helps identify the range in which a given inverter performs best.

2.10 Theoretical Review

The evaluation of solar inverters draws on established power-electronic theory, particularly DC-AC power conversion, MPPT, and Pulse Width Modulation (PWM) control (Teodorescu et al., 2011; Mohan et al., 2003).

2.10.1 DC-AC Power Conversion Theory

PV modules and batteries supply direct current, in which charge flows in one direction, whereas most appliances require alternating current, in which the direction changes periodically. Inverters achieve the conversion through semiconductor switches, such as IGBTs and MOSFETs, which rapidly switch the DC input on and off according to control signals to produce an alternating output waveform (Mohan et al., 2003). The efficiency of this process depends on switching frequency, semiconductor characteristics, and filtering, with some energy inevitably lost as conduction and switching losses. This theory is central to the present study because the ability to convert power efficiently under changing loads determines the usable energy delivered to appliances.

2.10.2 Maximum Power Point Tracking (MPPT) Theory

MPPT is a control technique that keeps a PV module operating at the point on its voltage-current curve where the product of voltage and current is highest, a point that shifts with solar radiation and temperature. Common techniques include the Perturb-and-Observe and Incremental Conductance methods, and more recent fuzzy-logic and artificial-intelligence approaches (Esram & Chapman, 2007). MPPT theory is relevant because load variation can influence the operating point of the source and, in turn, the ability of the system to sustain efficient conversion.

2.10.3 Pulse Width Modulation (PWM) Control Theory

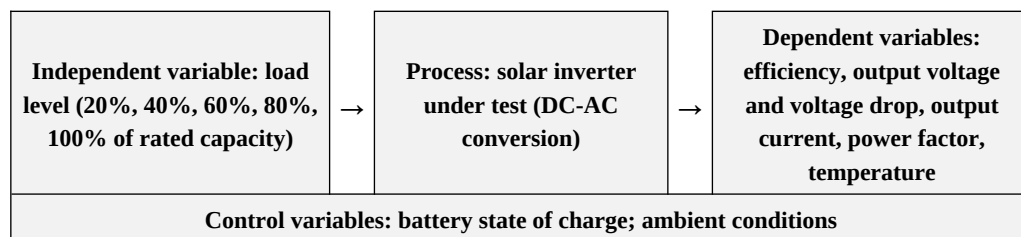
PWM controls an inverter's switches by varying the duration of switching pulses, producing voltage pulses that, once filtered, approximate a sinusoidal AC waveform; waveform quality depends on switching frequency, modulation index, and filter design (Mohan et al., 2003). PWM improves voltage regulation and reduces

harmonic distortion, allowing the inverter to respond to changing demand while maintaining acceptable output quality. Together, these three theories provide the framework for analysing efficiency, voltage stability, and output characteristics under variable load.

2.11 Conceptual Framework

Drawing on the performance parameters (Section 2.7) and the effect of load variation (Section 2.9), this study conceptualises the relationship between load level and inverter performance as an input-process-output system, illustrated in Figure 2.1. The electrical load level, expressed as a percentage of the inverter's rated capacity, is the independent variable, deliberately varied by the researcher. The inverter under test is the process stage, in which DC input power is converted to AC output power subject to internal losses. The dependent variables are conversion efficiency, AC output voltage (and the resulting voltage drop and regulation), AC output current, power factor, and case temperature; DC input voltage and current are the measured quantities from which input power is obtained. Battery state of charge and ambient conditions are control variables, held as constant as practicable so that changes in the dependent variables can be attributed to the variation in load level.

Figure 2.1: Conceptual Framework Showing the Relationship between Load Level and Inverter Performance



The control variables are held as constant as practicable during all runs.

This framework guides the experimental design in Chapter Three. By varying only the load level while holding the control variables as constant as practicable, any change observed in the dependent variables can reasonably be attributed to load variation, which is the relationship that underlies the hypotheses in Section 1.5.

2.12 Empirical Review

Table 2.2 summarises the empirical studies most relevant to this work. Chmielowiec et al. (2022) tested 26 new inverters (9 single-phase and 17 three-phase) available on the EU market, using a PV array simulator, a programmable AC source, and a power analyser arranged in accordance with EN 50530, and assessed conversion efficiency, static and dynamic MPPT efficiency, and harmonic emission at generation levels from 25% to 100% of rated power. Vaicys et al. (2022) combined a digital-twin simulation of a three-phase dual-system inverter with data from a conventional inverter. Krauter and Bendfeld (2024) ranked commercial microinverters using indoor efficiency tests and the Euro and CEC weighting schemes, and found that the ranking of individual models changed with the weighting used. Az Zahro et al. (2025) is the closest to the present work, since it examined a high-frequency inverter under resistive and mixed resistive-inductive load banks. Adesina et al. (2023) is a Nigerian case study of an off-grid inverter system.

Table 2.2: Summary of Reviewed Empirical Studies on Solar Inverter Performance

Author(s) and year	Focus of study	Key finding
Chmielowiec et al. (2022)	Laboratory tests of 26 commercial PV inverters using a PV simulator (EN 50530 set-up)	Efficiency indicators depended on the operating point (25%–100% of rated power) and on DC voltage level.
Vaicys et al. (2022)	Efficiency of a three-phase dual-system inverter for on-grid PV (simulation and comparison with conventional inverter data)	Weighted efficiency 0.22–0.27% higher than conventional inverters.
Krauter & Bendfeld (2024)	Euro and CEC efficiency ranking of commercial microinverters	Rankings changed with the weighting scheme used.
Az Zahro et al. (2025)	5 kW high-frequency inverter under resistive (0–1000 W) and resistive-inductive loads	Efficiency stable under resistive loads; sharp fluctuations with inductive loads.
Adesina et al. (2023)	3.5 kVA off-grid inverter system for a Nigerian household	Supplied an 11,260 Wh daily load for about 24 hours without interruption.

Collectively, these studies confirm that inverter efficiency and stability are not fixed values but depend on operating point, load type, and inverter design. However,

most concentrate on grid-connected systems, comparisons between inverter designs, or inductive and mixed load types. Few examine a single small-scale inverter step by step across a defined range of resistive load levels expressed as a percentage of its own rated capacity, which is the gap addressed by this study.

2.13 Summary of Literature Review and Research Gap

This chapter has shown that the solar inverter is one of the most critical components of a PV system, performing DC-AC conversion, voltage regulation, power-quality improvement, and protection (and MPPT in PV-connected designs), through types and topologies suited to different applications. It reviewed the key performance parameters, the international standards that govern their measurement, and, through theory, a conceptual framework, and recent empirical studies, showed that these parameters vary with load level, load type, and inverter design rather than remaining fixed at rated conditions.

Research Gap

Despite this body of work, comparatively few studies have evaluated a single small-scale inverter across a clearly defined range of resistive load levels expressed as a percentage of rated capacity. This is of practical relevance to the many small stand-alone and hybrid systems whose inverters operate under constantly changing demand. This study addresses the gap by experimentally evaluating a 1000 W solar inverter at five load levels (20%, 40%, 60%, 80%, and 100% of rated capacity), using controlled resistive loads and standard electrical measuring instruments, as detailed in Chapter Three.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the methodology used for the experimental evaluation of a solar inverter under variable load conditions. It describes the research design, the materials and equipment, the experimental set-up, the procedure, the method of data collection, and the relations used for data analysis, and it outlines the precautions taken to ensure the validity and reliability of the results. The study did not involve the design or construction of a new inverter; it evaluated the performance of an existing, commercially available inverter subjected to a range of controlled electrical loads.

3.2 Research Design

The study adopted a quantitative, experimental research design. This was considered most appropriate because the study required the direct manipulation of an independent variable (electrical load level) under controlled conditions and the measurement of its effect on a set of dependent variables.

The independent variable is the electrical load level, expressed as a percentage of the inverter's rated capacity (20%, 40%, 60%, 80%, and 100%). The dependent variables are conversion efficiency, AC output voltage (and the resulting voltage drop and regulation), AC output current, power factor, and inverter case temperature; DC input voltage and current were measured so that DC input power and efficiency could be calculated. All experiments were conducted under controlled laboratory conditions, with battery state of charge and ambient conditions held as constant as practicable in order to isolate the effect of load variation.

3.3 Description of the System under Study

The equipment under test is a single-phase solar inverter with a rated output power of 1000 W, selected on the basis of availability. It is powered from a rechargeable

12 V deep-cycle battery, which was charged from a battery charger or a PV charging system. The inverter converts the DC power from the battery into single-phase AC power at the nominal local supply voltage of 230 V, which is supplied to a controlled variable load.

A controlled resistive load bank was used instead of randomly selected domestic appliances. Controlled resistive loads (such as calibrated lamp banks or resistive heating elements) can be adjusted to a precisely known value, expressed as a percentage of the inverter's rated capacity, which improves the safety, consistency, and repeatability of the experiment. Because the loads were resistive and the power factor was close to unity, real and apparent power were practically equal, so whether the inverter's rating is read in watts or volt-amperes does not materially affect the load levels used.

3.4 Materials and Equipment

The equipment and materials used are summarised in Table 3.1. They were selected for accurate voltage and current measurement, the use of controlled loads, temperature monitoring, and systematic data recording.

Table 3.1: List of Equipment/Instruments Used and Their Functions

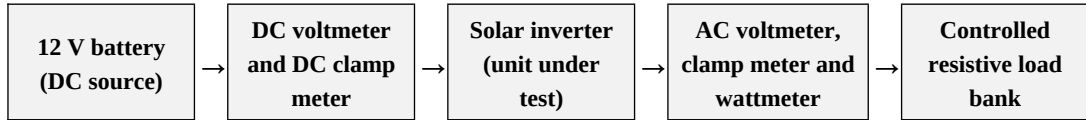
Equipment/Material	Function in the experiment
Solar inverter (1000 W rated)	Device under test; converts DC battery power to AC output
Rechargeable 12 V deep-cycle battery	Provides a stable, known DC power source to the inverter
Battery charger / PV charging system	Restores the battery to the same state of charge before each run
Digital multimeter	Measures DC input voltage and AC output voltage
Digital clamp meter (DC and AC)	Measures DC input current (up to about 95 A) and AC output current
AC wattmeter / power meter	Measures real AC output power and power factor
Controlled resistive load bank	Provides adjustable, repeatable load levels (percentage of rated capacity)
Thermometer / non-contact temperature sensor	Monitors inverter case/heat-sink temperature during operation
Stopwatch / digital timer	Times the stabilisation period at each load level

Equipment/Material	Function in the experiment
Connecting wires, switches, fuses/circuit breakers	Circuit connection and overload/short-circuit protection
Data-recording sheet / laptop computer	Recording and analysis of experimental readings

3.5 Experimental Set-Up

The experimental arrangement is shown in Figure 3.1. The battery (DC source) is connected to the DC input terminals of the inverter, with a clamp meter measuring the DC input current and a voltmeter connected in parallel across the input terminals measuring the input voltage. The DC current at full load is about 95 A, which is beyond the range of ordinary multimeter current inputs, so a DC clamp meter was used. On the output side, the AC terminals of the inverter feed the controlled resistive load bank, with a clamp meter measuring the output current, a voltmeter connected in parallel across the output, and a wattmeter measuring the real power delivered to the load. A temperature sensor was positioned on the inverter's heat sink/casing to monitor thermal behaviour during each run.

Figure 3.1: Block Diagram of the Experimental Set-Up



The temperature sensor on the inverter case is not shown in the signal path.

3.6 Determination of Load Levels

So that the results are reproducible, load levels were not described in vague terms such as “low”, “medium”, and “high”. Each level was defined as a percentage of the inverter's rated output capacity using Equation 3.1:

$$\text{Load (\%)} = (P_L \div P_R) \times 100 \quad (3.1)$$

where P_L is the actual power drawn by the resistive load bank at a given setting and P_R is the manufacturer's rated output power of the inverter (1000 W). Five load levels were adopted (Table 3.2) to cover light loading, the medium-load region in which peak efficiency is typically expected, and full rated loading.

Table 3.2: Load Levels as a Percentage of the 1000 W Rated Inverter Capacity

Load level	Percentage of rated capacity	Load power	Loading description
1	20%	200 W	Light load
2	40%	400 W	Light–medium load
3	60%	600 W	Medium load
4	80%	800 W	Medium–heavy load
5	100%	1000 W	Full (rated) load

3.7 Experimental Procedure and Controls

The following controls were applied throughout the experiment to minimise extraneous variation, so that observed changes in performance could be attributed to load variation:

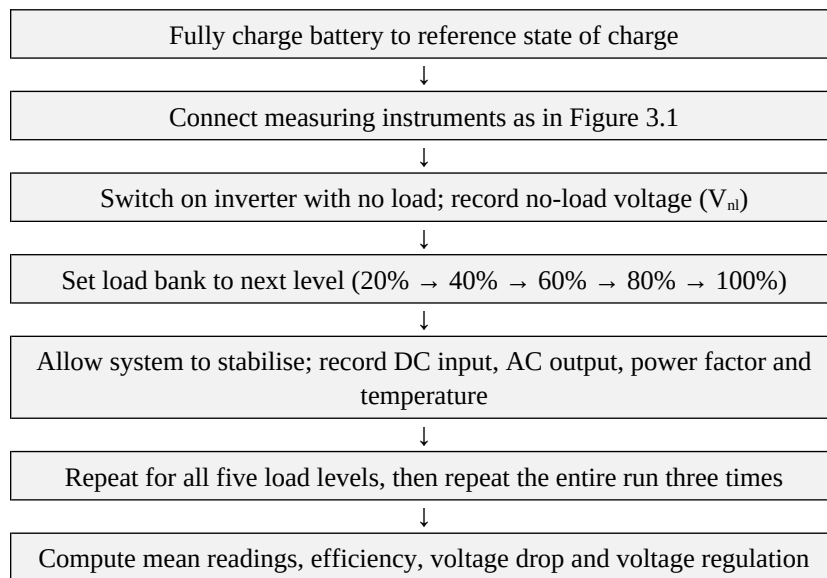
1. **Same battery throughout:** the same battery was used for every run, so that differences between batteries could not be mistaken for differences in inverter performance.
2. **Consistent initial state of charge:** the battery was fully charged to the same state before each run, using the same charging procedure.
3. **Predetermined load levels:** the load bank was set to the exact percentages in Table 3.2.
4. **Consistent ambient conditions:** all tests were conducted indoors, at the same location and under similar ambient temperature, as far as practicable.

The experiment was then carried out in the following sequence:

1. The battery was fully charged and allowed to rest, and its open-circuit voltage was recorded.
2. The instruments were connected as shown in Figure 3.1, and all connections were checked for correctness and safety before power was applied.
3. The inverter was switched on with no load connected, and the no-load output voltage (V_{nl}) was recorded.
4. The load bank was set to the first load level (20% of rated capacity), and the system was allowed to stabilise for a fixed period, timed with a stopwatch.
5. At the stabilised condition, the DC input voltage and current, AC output voltage and current, AC output power, power factor, and inverter temperature were read and recorded on the data-recording sheet (Table 3.3).

6. The load was then increased in ascending order to 40%, 60%, 80%, and 100% of rated capacity, repeating step 5 at each level, with the battery and other conditions kept as constant as practicable.
7. The full-load output voltage (V_{fl}), obtained at the 100% load level, was recorded for the voltage-regulation calculation.
8. The complete run (no load through to 100% load) was repeated three times, with the battery recharged to the same state of charge between runs, and the mean of the readings at each load level was used for analysis.
9. On completion, all equipment was switched off and disconnected.

Figure 3.2: Flowchart of the Experimental Procedure



3.8 Method of Data Collection

Data were collected by direct instrument readings at each of the five load levels in Section 3.6. The parameters recorded were DC input voltage (V), DC input current (A), AC output voltage (V), AC output current (A), AC output (real) power (W), power factor, and inverter case temperature (°C). Readings were entered on a structured data-recording sheet, the format of which is shown in Table 3.3, and then transferred to a computer spreadsheet for computation and graphical analysis. The mean values obtained from the three repeated runs are presented in Chapter Four.

Table 3.3: Format of the Data Recording Sheet

Load (%)	V _{in} (V)	I _{in} (A)	V _{out} (V)	I _{out} (A)	P _{out} (W)	PF	Temp. (°C)
No load							
20							
40							
60							
80							
100							

3.9 Method of Data Analysis

The mean values from the three runs were analysed using descriptive statistics (means, differences, and percentage values) and standard power-electronic relations. The DC input power at each load level was calculated from the measured DC voltage and current using Equation 3.2:

$$\text{DC Input Power, } P_{in} \text{ (W)} = V_{in} \times I_{in} \quad (3.2)$$

The inverter efficiency at each load level was then calculated using Equation 3.3 (the same relation as Equation 2.1):

$$\text{Efficiency, } \eta \text{ (\%)} = (P_{out} \div P_{in}) \times 100 \quad (3.3)$$

The voltage drop at each load level, and the voltage regulation at full load, were calculated from the no-load and loaded output voltages using Equation 3.4 (the same relation as Equation 2.2, with V_L the output voltage at the load level considered; at 100% load, V_L = V_{fl} and the result is the voltage regulation):

$$\text{VD or VR (\%)} = [(V_{nl} - V_L) \div V_L] \times 100 \quad (3.4)$$

The power factor was read from the wattmeter and checked against Equation 3.5:

$$\text{Power Factor, PF} = P_{out} \text{ (W)} \div [V_{out} \times I_{out}] \text{ (VA)} \quad (3.5)$$

The power loss in the inverter at each level was obtained as $P_{loss} = P_{in} - P_{out}$. The calculated values were tabulated and plotted with percentage load on the horizontal axis

to show the trends of efficiency, output voltage, output current, and temperature against load, and these trends form the basis of the discussion in Chapter Four.

Decision rule for the hypotheses. Because the analysis is based on the mean of the repeated runs at each load level, each null hypothesis in Section 1.5 was assessed by comparing the change in the relevant parameter across the load levels with the measurement uncertainty of the instruments, and by checking whether the change followed a consistent trend over all five levels. A null hypothesis was rejected in favour of its alternative when the observed change was clearly larger than the instrument uncertainty and followed a consistent trend; otherwise it was retained. The conclusions apply to the inverter tested.

3.10 Validity, Reliability and Precautions

Several precautions were taken to ensure valid and reliable results. First, the measuring instruments were checked and, where possible, cross-verified against one another before use, to reduce instrument-related error. Second, readings were taken only after the system had stabilised at each load level, to avoid recording transient values. Third, each complete run was repeated three times and the mean values used, which reduces the effect of random measurement error. Fourth, all connections were secured and fuses and circuit breakers were installed to protect the equipment and the researcher from short circuits and overloads.

The same instruments were used at every load level, so any systematic instrument error affects all levels in a similar way; trends across load levels are therefore more reliable than the absolute values. Internal validity was further supported by holding battery state of charge and ambient conditions as constant as practicable across all runs.

3.11 Limitations of the Methodology

Notwithstanding these precautions, the methodology has limitations in addition to those in Section 1.8. The accuracy of the calculated efficiency and voltage-regulation

values depends on the precision and calibration of the instruments, and small instrument errors may accumulate in derived quantities. The loads were purely resistive, so behaviour under inductive, capacitive, or non-linear loads was outside the scope of the equipment available. The load levels were applied in ascending order, so heating of the inverter and the gradual drain on the battery accumulate along the sequence, and their effects could not be separated from the effect of load itself; the battery was recharged between runs to limit this. Because the analysis used the mean of the repeated runs, inferential tests such as analysis of variance, which need the individual replicate readings, were not applied. Finally, the tests were of short duration under laboratory conditions, so long-term effects of continuous operation, ageing, and outdoor exposure were not assessed.

3.12 Summary

This chapter described the experimental, quantitative design used to evaluate a 1000 W solar inverter under variable load conditions. It presented the equipment, the set-up, the procedure and controls, the data collection method, and the relations used for analysis, including the decision rule for the hypotheses. The results obtained with this methodology are presented and discussed in Chapter Four.

CHAPTER FOUR

RESULTS, PRESENTATION AND DISCUSSION OF FINDINGS

4.1 Introduction

This chapter presents the results of the experimental evaluation of the 1000 W solar inverter described in Chapter Three. It presents the averaged readings, the calculated performance parameters (efficiency, power loss, voltage drop and regulation, and power factor), a graphical analysis of the trends, a discussion of the findings in relation to the literature in Chapter Two, and the test of the hypotheses stated in Section 1.5.

4.2 Presentation of Results

Table 4.1 presents the mean of the three repeated readings at each of the five load levels in Section 3.6, together with the no-load reference voltage recorded before loading began.

Table 4.1: Mean Experimental Readings at Each Load Level

Load (%)	V _{in} (V)	I _{in} (A)	P _{in} (W)	V _{out} (V)	I _{out} (A)	P _{out} (W)	Temp. (°C)
No load	12.8	—	—	232.0	—	—	28
20	12.6	19.36	243.9	230.5	0.87	200.0	31
40	12.5	35.55	444.4	229.8	1.74	400.0	35
60	12.3	52.45	645.1	228.9	2.62	600.0	40
80	12.1	71.87	869.6	227.6	3.52	800.0	46
100	11.9	94.42	1123.6	225.8	4.43	1000.0	53

The DC input power at each level was calculated using Equation 3.2, and the efficiency, voltage drop, and power loss using Equations 3.3, 3.4, and $P_{\text{loss}} = P_{\text{in}} - P_{\text{out}}$. As a worked example, at the 60% load level:

$$P_{\text{in}} = V_{\text{in}} \times I_{\text{in}} = 12.3 \times 52.45 = 645.1 \text{ W}$$

$$\eta (\%) = (P_{\text{out}} \div P_{\text{in}}) \times 100 = (600.0 \div 645.1) \times 100 = 93.0\%$$

The calculated parameters for all five load levels are presented in Table 4.2.

Table 4.2: Calculated Performance Parameters at Each Load Level

Load (%)	Efficiency, η (%)	Power loss (W)	Voltage drop, VD (%)	Power factor, PF
20	82.0	43.9	0.65	0.99
40	90.0	44.4	0.96	1.00
60	93.0	45.1	1.35	1.00
80	92.0	69.6	1.93	0.99
100	89.0	123.6	2.75	0.99

Voltage drop is calculated from the no-load voltage of 232.0 V (Equation 3.4); at 100% load it equals the voltage regulation of the inverter, VR = 2.75%.

The simple (unweighted) mean of the five efficiency values is 89.2%, which is 3.8 percentage points below the peak of 93.0%. Efficiency at rated load (89.0%) is 4.0 points below the peak, and efficiency at 20% load (82.0%) is 11.0 points below it. A single peak figure would therefore overstate the efficiency of the inverter at the lightest and the heaviest loads tested, which is the situation described in the problem statement (Section 1.2). The output voltage stayed between 225.8 V and 232.0 V throughout, which is within -1.8% and $+0.9\%$ of the nominal 230 V.

4.3 Graphical Analysis of Results

The trends in Tables 4.1 and 4.2 are shown in Figures 4.1 to 4.4 to make the relationship between load level and each parameter easier to see.

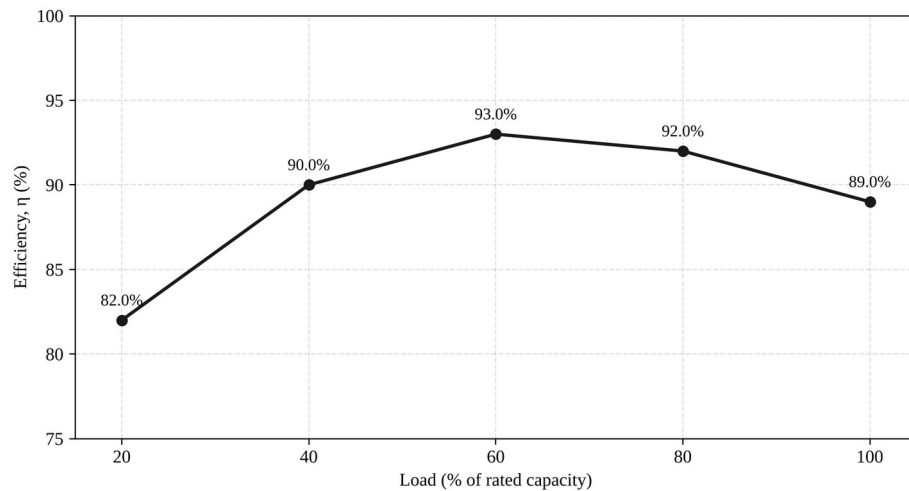


Figure 4.1: Inverter Efficiency versus Load Level

Figure 4.1 shows the non-linear relationship between load level and efficiency described in Section 2.9. Efficiency is lowest at 20% load (82.0%), rises to a peak at 60% load (93.0%), and then declines at 80% and 100% load (92.0% and 89.0%). The power loss in Table 4.2 helps to explain this shape. The loss is almost constant at 44–45 W from 20% to 60% load, so at 20% load this fixed loss is a large share of the 243.9 W drawn, which gives the low efficiency. Above 60% load the loss rises to 69.6 W and then 123.6 W as the DC current increases to 71.87 A and 94.42 A, so the increase in current-dependent losses lowers the efficiency again. This is consistent with the loss mechanisms described in Section 2.7 (King et al., 2007; Mohan et al., 2003).

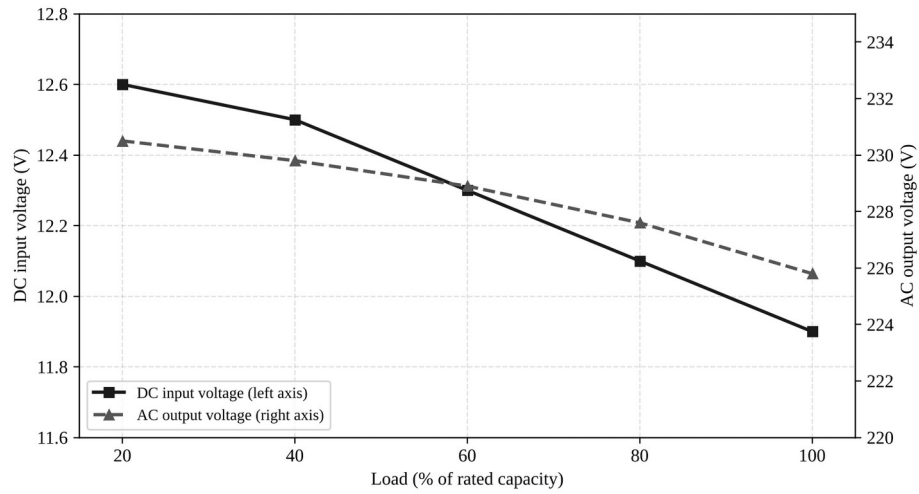


Figure 4.2: DC Input Voltage and AC Output Voltage versus Load Level

Figure 4.2 shows that both voltages decline steadily as load increases, from 12.6 V to 11.9 V on the DC side and from 230.5 V to 225.8 V on the AC side between 20% and 100% load. Dividing the fall of the DC voltage below its open-circuit value of 12.8 V by the DC current gives an effective resistance of 8.4–10.3 m Ω at every level, which suggests that the DC-side droop is mainly ohmic, consistent with the internal resistance of the battery and the connections. The AC-side droop reflects the inverter's own voltage regulation; the voltage regulation of 2.75% (Table 4.2) indicates a well-regulated output for the inverter tested.

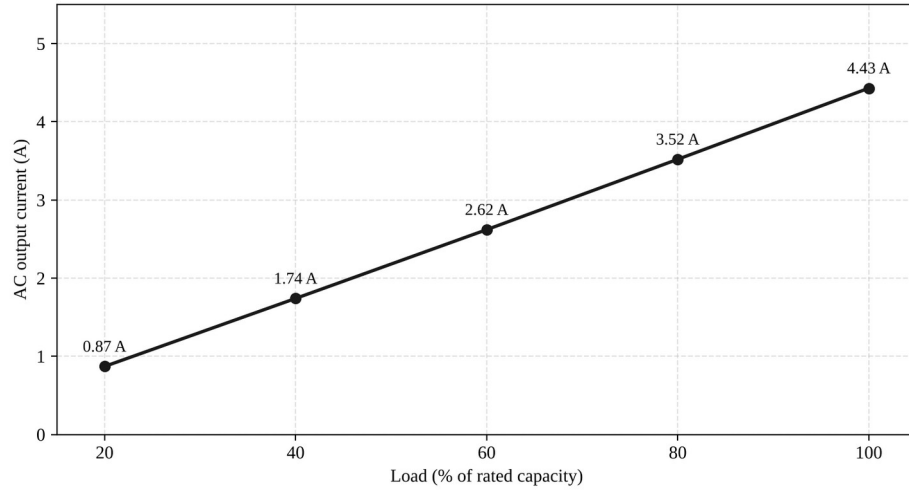


Figure 4.3: AC Output Current versus Load Level

Figure 4.3 shows that the AC output current rises almost in proportion to the load, from 0.87 A at 20% load to 4.43 A at 100% load, a factor of 5.1 for a five-fold increase in load. This is the expected behaviour for a resistive load at a nearly constant voltage and a power factor close to unity, since the current is then approximately $P_{\text{out}} \div V_{\text{out}}$. The power factor remained between 0.99 and 1.00 at every level (Table 4.2), consistent with the resistive load bank used.

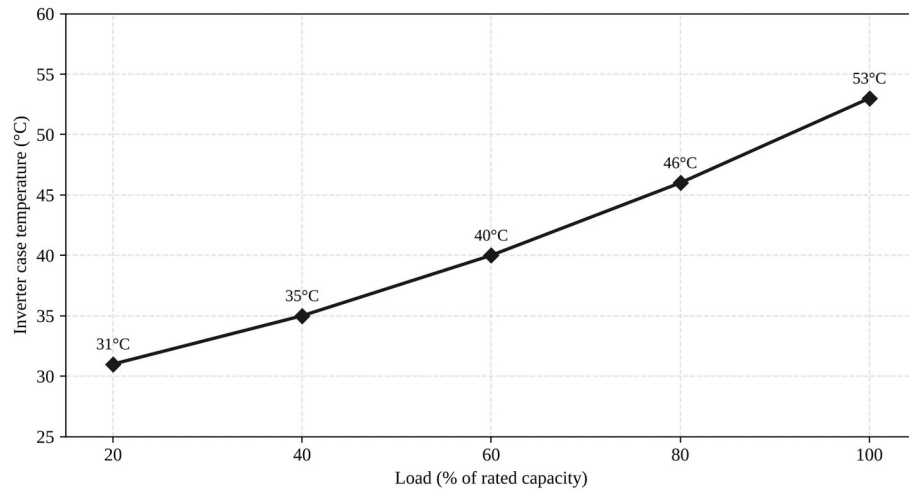


Figure 4.4: Inverter Case Temperature versus Load Level

Figure 4.4 shows that the case temperature rises steadily with load, from 31°C at 20% load to 53°C at 100% load, compared with 28°C at no load, and that the largest step (7°C) occurs between 80% and 100% load, where the power loss also rises most. The

temperature continued to rise between 20% and 60% load even though the loss was almost constant, which indicates that the readings also reflect heat accumulating in the inverter along the ascending test sequence (Section 3.11). The temperature readings should therefore be read as the thermal behaviour of this test sequence rather than as steady-state temperatures at each load.

4.4 Discussion of Findings

The efficiency pattern in Figure 4.1, low at light load, highest at a mid-range load, and lower again at rated load, is consistent with the load-dependent efficiency reported for commercial inverters (Chmielowiec et al., 2022; King et al., 2007) and with the loss mechanisms in Section 2.9. The result also shows that a single efficiency figure does not describe the inverter: using only the peak value would overstate efficiency by 4.0 points at rated load and by 11.0 points at 20% load. This supports the use of several load levels, and of weighted measures such as the Euro and CEC efficiencies (Krauter & Bendfeld, 2024), that was argued in Section 2.7.

Az Zahro et al. (2025) reported stable efficiency for a 5 kW inverter under resistive loads of up to 1000 W, which is about 20% of that inverter's rating. The present study covers 20% to 100% of the rating of a 1000 W inverter and found that efficiency varied with load even under purely resistive loading. The two studies cover different parts of the relative load range and different inverters, so they are not directly comparable, but both indicate that the magnitude of the load relative to the rating is an important factor.

The AC output voltage decreased by 4.7 V (2.0%) between 20% and 100% load, and by 6.2 V from the no-load value, giving a voltage regulation of 2.75%. As a reference only, the European standard for public low-voltage supply allows the voltage to vary by $\pm 10\%$ about the nominal value (CENELEC, 2010), and the output of the inverter remained well inside that band at every load level. Because the same instruments were used at every level, the consistent downward trend is more reliable

than the small absolute change, which is of the same order as the accuracy of ordinary digital meters.

The temperature rise with load (Figure 4.4) agrees with the expectation in Sections 2.7 and 2.9 that heavier loading increases internal power dissipation and heat generation. Efficiency began to fall at the load levels (80% and 100%) at which the power loss and the temperature rose most, so a thermal contribution to the lower efficiency at high load is possible. However, the study did not separate conduction, switching, and thermal effects, and the more direct explanation supported by the data is the increase in current-dependent losses with load.

4.5 Test of Hypotheses

The three null hypotheses in Section 1.5 were assessed using the decision rule in Section 3.9, which compares the observed change in each parameter across the load levels with the instrument uncertainty and checks for a consistent trend.

1. **H₀₁ (no difference in efficiency across load levels):** the efficiency ranged from 82.0% at 20% load to 93.0% at 60% load, a spread of 11.0 percentage points (about 13% of the lowest value), and followed a clear rise-and-fall pattern. This change is far larger than the uncertainty expected from the instruments used, so H₀₁ is rejected and H₁₁ is accepted for the inverter tested.
2. **H₀₂ (load has no effect on output voltage):** the AC output voltage fell at every step, from 230.5 V at 20% load to 225.8 V at 100% load (and from 232.0 V at no load), a consistent monotonic trend. H₀₂ is therefore rejected and H₁₂ accepted, although the size of the effect is small (about 2%) and the output stayed within a narrow band.
3. **H₀₃ (no relationship between load and overall performance):** efficiency showed a clear non-linear relationship with load (Figure 4.1), and the voltage drop increased steadily with load from 0.65% to 2.75% (Table 4.2). H₀₃ is therefore rejected and H₁₃ accepted.

These conclusions rest on the mean values of the repeated runs and apply to the inverter tested; they were not confirmed by an inferential test on the individual readings (Section 3.11).

4.6 Summary

This chapter presented the results for the inverter at five load levels. Efficiency was lowest at 20% load (82.0%), peaked at 60% load (93.0%), and fell to 89.0% at 100% load, with a nearly constant loss of 44–45 W up to 60% load and a rapid increase in loss above it. The output voltage fell steadily with load, giving a voltage regulation of 2.75%; the output current rose almost in proportion to the load; the power factor stayed between 0.99 and 1.00; and the case temperature rose from 31°C to 53°C. These findings are consistent with the literature in Chapter Two, and under the decision rule in Section 3.9 all three null hypotheses were rejected. The conclusions and recommendations are presented in Chapter Five.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents a summary of the study, the findings, the conclusions drawn from them, the recommendations, the contribution to knowledge, and suggestions for further research.

5.2 Summary of the Study

This study set out to evaluate the performance of a solar inverter under variable load conditions. Chapter One introduced the background, stated the problem that inverter performance is commonly specified only at peak or rated conditions although inverters operate under changing demand, and set out the aim, objectives, research questions, and hypotheses. Chapter Two reviewed literature on PV systems, inverter types, topologies, performance parameters, testing standards, the effect of variable load, and previous empirical studies, and identified a gap in the step-wise evaluation of a single small-scale inverter across a defined range of resistive load levels. Chapter Three described the experimental design, the equipment, the procedure, and the analysis, in which a 1000 W inverter was tested at 20%, 40%, 60%, 80%, and 100% of its rated capacity. Chapter Four presented and discussed the results.

5.3 Summary of Findings

The key findings for the inverter tested are as follows:

1. Efficiency was not constant across the load range: it was lowest at light load (82.0% at 20% load), peaked at a mid-range load (93.0% at 60% load), and fell to 89.0% at rated load. The simple mean of the five values was 89.2%.
2. The power loss was almost constant at 44–45 W from 20% to 60% load and then rose to 69.6 W at 80% load and 123.6 W at 100% load, which accounts for the low efficiency at light load and the fall in efficiency near rated load.

3. The AC output voltage fell steadily with load, from 230.5 V at 20% load to 225.8 V at 100% load, giving a voltage regulation of 2.75% between no-load and full-load conditions; the output stayed within about 2% of the nominal 230 V.
4. The AC output current rose almost in proportion to the load, from 0.87 A to 4.43 A, and the power factor remained between 0.99 and 1.00, consistent with a resistive load.
5. The case temperature rose steadily from 31°C at 20% load to 53°C at 100% load, and the largest increase occurred where the power loss also rose most.
6. Under the decision rule in Section 3.9, all three null hypotheses in Section 1.5 were rejected, indicating that load level has a measurable effect on the efficiency, output voltage, and overall performance of the inverter tested.

Table 5.1: Objectives of the Study and the Corresponding Findings

Objective	Corresponding finding
1. Determine the input and output power at different load levels	Input power rose from 243.9 W to 1123.6 W for an output of 200 W to 1000 W (Table 4.1).
2. Evaluate the efficiency at varying loads	Efficiency was 82.0% to 93.0%, with the peak at 60% load (Figure 4.1).
3. Examine the effect of load on output voltage, current, and power factor	Voltage fell by 4.7 V, current rose in proportion to load, and the power factor stayed at 0.99–1.00 (Figures 4.2 and 4.3).
4. Assess the operational stability (voltage regulation and temperature)	Voltage regulation was 2.75%, and the case temperature rose from 31°C to 53°C (Figure 4.4).
5. Identify the relationship between load demand and overall performance	Efficiency varied non-linearly with load and voltage drop increased steadily with load; all null hypotheses were rejected (Section 4.5).

5.4 Conclusion

It is concluded that the performance of the 1000 W solar inverter evaluated is clearly influenced by the electrical load under which it operates. The inverter does not perform uniformly across its operating range: it is less efficient at light load, reaches its best efficiency in the mid-range loading region, and shows a decline in efficiency, with rising temperature, as it approaches full rated capacity. Output voltage regulation was good across the tested range but degraded progressively as load increased. These findings, for the inverter tested, confirm that a single efficiency figure quoted at peak or

rated conditions is not sufficient to describe real-world performance, and that evaluation across a range of load levels gives a more complete and realistic picture of inverter behaviour for practical PV applications.

5.5 Recommendations

Based on the results for the inverter tested, the following recommendations are made:

1. Where possible, PV systems should be sized so that the inverter normally operates in the mid-range load region, about 40% to 80% of rated capacity for the unit tested, where efficiency was 90% or higher, rather than at very light or very heavy loads.
2. Adequate cooling and ventilation should be provided for inverters expected to operate frequently at or near full rated capacity, given the rise in temperature and power loss at higher loads.
3. Manufacturers and system designers should report, or request, weighted efficiency figures (such as the Euro or CEC efficiency in Section 2.7), or efficiency at several load levels, rather than a single peak value.
4. Users and installers of small-scale inverters should periodically monitor output voltage and temperature, particularly where load demand is expected to approach the rated capacity for extended periods.

5.6 Contribution to Knowledge

This study provides measured data on the efficiency, power loss, output voltage, current, power factor, and temperature of a 1000 W stand-alone solar inverter at five load levels expressed as percentages of its rated capacity, a step-wise evaluation of a single small-scale inverter that the reviewed literature addresses only in part. It also shows, using the measured power loss, how a nearly constant loss at light load and a rapidly increasing loss near rated load together give the rise-and-fall efficiency curve, and it provides a clear and repeatable procedure that students and practitioners can use to evaluate other small inverters.

5.7 Suggestions for Further Studies

On the basis of the limitations in Sections 1.8 and 3.11, the following areas are suggested for further research:

1. Evaluating inverter performance under inductive, capacitive, and non-linear loads, in addition to the resistive loads used here, since many real loads are not purely resistive.
2. Repeating the tests with the load levels applied in randomised order or with cooling intervals, and analysing the individual replicate readings with analysis of variance, so that the effect of load can be separated from heating and battery effects and tested statistically.
3. Comparing several inverters of different topologies or switching devices under the same variable-load procedure, and comparing the measured efficiency curve with the manufacturers' stated values.
4. Conducting long-duration or outdoor field tests to assess how ageing, continuous operation, and real environmental conditions affect the load-dependent trends observed under laboratory conditions.

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